Nibraas Khan

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Cambridge, MA - USA

EDUCATION

- **Vanderbilt University** — *Aug 2020 - July 2026*
  Ph.D. Computer Science — Nashville, TN
  - **Dissertation:** *Advancing Ecologically Valid Human Motion Analysis Through Wearable Sensing and Machine Learning.*
- **Middle Tennessee State University** — *Aug 2017 - May 2020*
  B.S. Computer Science — Murfreesboro, TN

EXPERIENCE

- **Cambridge Mobile Telematics** — *Sep 2026 - Present*
  Machine Learning Engineer 2 — Cambridge, MA
  - Working on the foundation model team.
- **Mitsubishi Electric Research Laboratories** — *Jan 2026 - July 2026*
  Research Scientist Intern — Cambridge, MA
  - Built pretraining and post-training recipes for on-device vision-language-action models enabling at-home human-robot interaction, fusing vision, proprioception, affective signals, and WiFi-CSI scene maps.
  - Developed a PPO-based framework for adaptive memory and context management across Neural Processes and foundation models under bounded compute; 17% improvement over heuristic baselines.
- **Actual Insight** — *Oct 2025 - Jan 2026*
  Applied Scientist Intern - Multi-Agent Systems — Nashville, TN
  - Designed and deployed multi-agent LLM pipelines (LangGraph) that turned 15+ enterprise clients' data into actionable insights with 100% output-to-source provenance, role-based scoping, and PII redaction.
  - Built a LangChain natural-language-to-SQL workflow that let business users query their data in plain English, with a locality-sensitive-hashing cache that cut prompt-token usage by 18%.
- **Vanderbilt University - Robotics and Autonomous Systems Lab** — *May 2020 - Present*
  Graduate Researcher — Nashville, TN
  - Engineered a full-stack IMU/ESP32 boxing system, driven by a 30M-parameter PyTorch transformer fine-tuned for movement-quality assessment and validated by 17 athletes and 6 coahces.
  - Built REACT, an explainable few-shot early-agitation detector trained on 47.5 hours of multimodal wearable-and-audio data (84.6% F1) and deployed with 15 children in clinical settings over the past year.
  - Made predictions interpretable with three model-agnostic algorithms (ablation, counterfactuals, and knowledge-graph-grounded LLM explanations) that surface the movement patterns behind each score.

SELECTED PUBLICATIONS

- **Khan, N.**, Cole, K., Sarkar, N. 2026. *Toward Assessing Functional Decline in Mild Cognitive Impairment Using Wearable Sensors and Explainable Machine Learning*. Digital Health.
- **Khan, N.**, Wichern, G., Laughman, C. R. 2026. *Reinforced Neural Processes: Memory-Efficient Time-Series Forecasting with a World-Feedback-Trained Memory Policy*. ICML 2026 RLxF Workshop.

SKILLS

- **Foundation Models & LLMs:** Pretraining, Fine-Tuning, RAG, Embeddings, Agentic/Multi-Agent (LangGraph), VLMs/VLA
- **ML Methods:** Reinforcement Learning (PPO), Explainable AI, Few-Shot & Meta-Learning, Time-Series, Deep Learning (Transformers, RNNs, CNNs), Computer Vision
- **Systems & Hardware:** Embedded firmware (ESP32), IMU & multimodal sensor integration, real-time on-device inference, full-stack (React, TypeScript)
- **Frameworks & MLOps:** PyTorch, TensorFlow/Keras, Hugging Face Transformers, Scikit-learn, AWS (EC2, S3, SageMaker), Docker, Kafka, Airflow, Git
- **Languages:** Python, C#, Kotlin, TypeScript, SQL, Bash